1 **(a)** Define gravitational potential at a point.

....

....

.... [2]

(b) A uniform spherical planet has radius R. The gravitational potential at its surface is $-\Phi$.

P is a point on a line that passes through the centre of the planet, at variable displacement x from the centre, as shown in Figure 1.1.

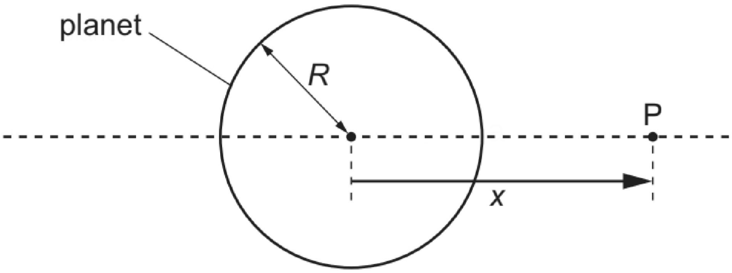

Figure 1.1

Figure 1.2 shows the variation with x of the gravitational potential ϕ due to the planet in the region outside the planet close to its surface.

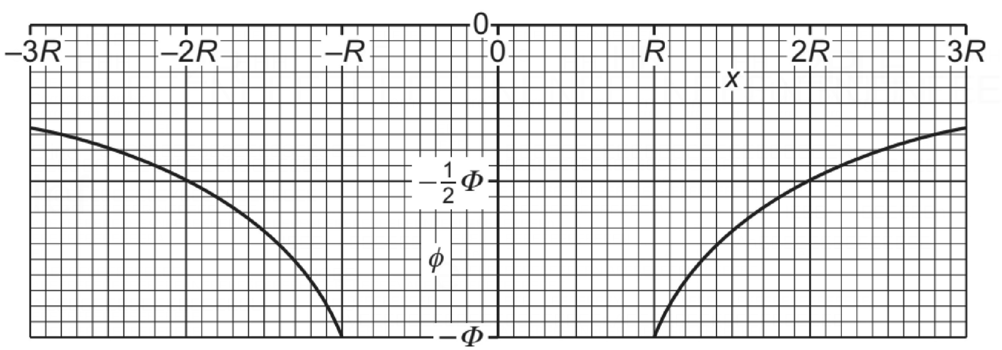

Figure 1.2

(i) Explain why ϕ is negative for all values of x.

....

....

.... [2]

(ii) State an expression, in terms of Φ and R, for the mass M of the planet. Identify any other symbols that you use.

M = ... [2]

(iii) Determine an expression, in terms of Φ and R, for the gravitational field strength g_0 at the surface of the planet.

g_0 = ... [1]

(iv) On Figure 1.3, sketch the variation of the gravitational field g with x between $x = -3R$ and $x = 3R$. Do **not** include the region inside the sphere between $x = -R$ and $x = R$.

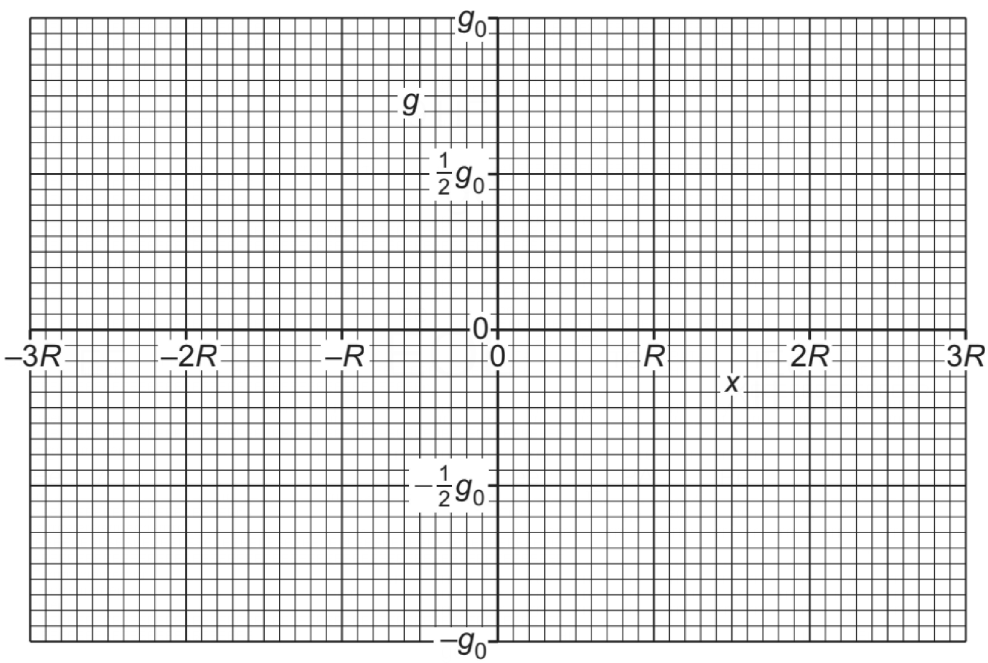

Figure 1.3

[3]

[Total: 10]